Describing Wood-Bat Success: The Northwoods League Scout Score

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Abstract

Transitioning from metal to wood bats is a critical hurdle for collegiate prospects, as metal often masks mechanical flaws and inflates traditional production. This study utilizes 2025 Northwoods League data to identify which collegiate metrics best predict success with wood bats. Using Random-Forest Regression and correlation analysis, we developed a "Scout Score" that weighs process-oriented metrics—specifically Isolated Power (ISO) and Plate Discipline (BB/K)—against "flashy" traditional stats like average, home runs, and RBIs to better identify translatable professional tools.

Our Proprietary Model: The Scout Score

Scout Score: Synthetic score designed to reward **PROCESS** over results.

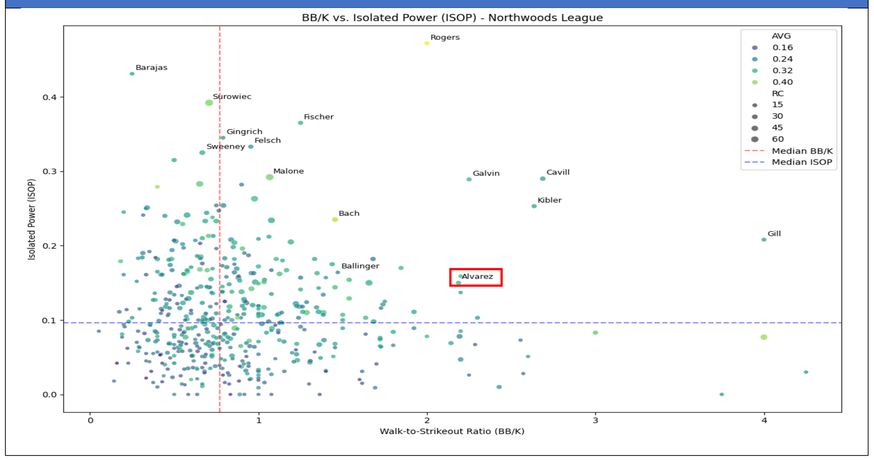
SCOUT SCORE COMPOSITION

ISOP	BB%	K%	AVG
40%	30%	20%	10%

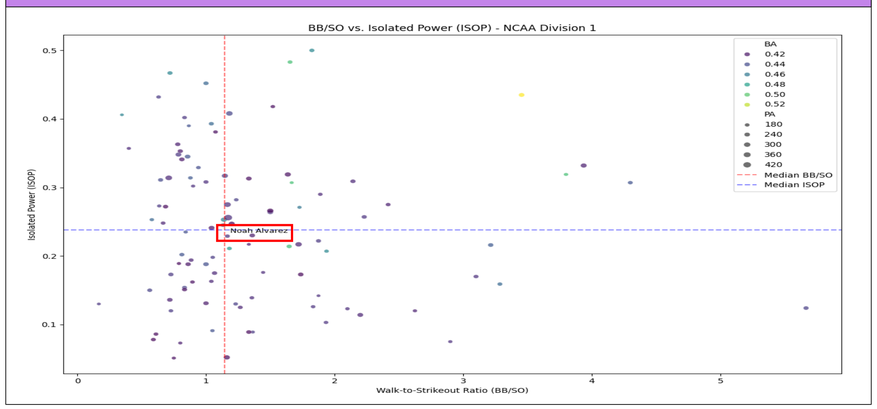
Naïve Approach: Runs Created (RC)

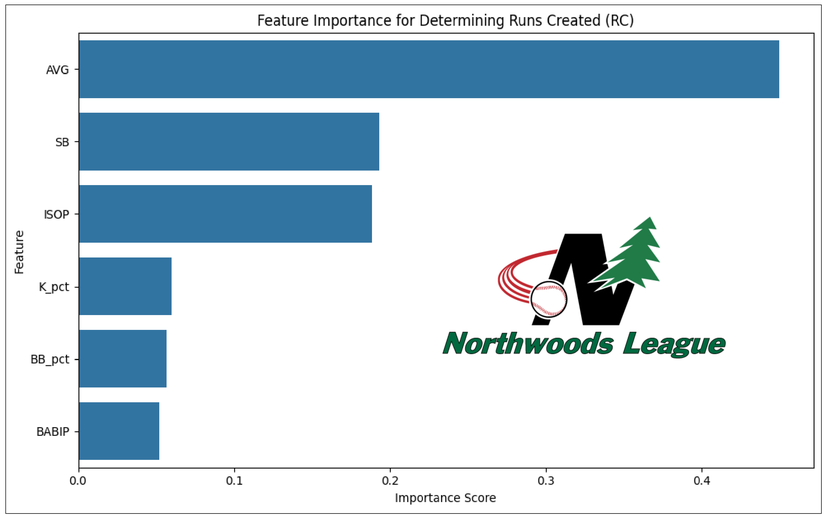
Player	Team	AVG	RC	Scout_Score
Ethan Surowiec	DUL	.387	73.70	.398
Noah Malone	WAU	.381	68.00	.375
Dylan Schlotterback	WAU	.318	55.86	.319
Henry Allen	MOT	.366	55.49	.349
Max Soliz	WAU	.338	54.05	.355

Northwoods League | Walk-to-Strikeout Ratio vs. Isolated Power



NCAA Division 1 | Walk-to-Strikeout Ratio vs. Isolated Power





Metric	Full Name	Definition / Formula
RC	Runs Created	Formulaic estimate of total runs contributed to the team
ISOP	Isolated Power	Measures raw power by excluding singles (SLG - AVG)
BB%	Walk Percentage	Ratio of walks to total AB's
K%	Strikeout Percentage	Ratio of strikeouts to total AB's
AVG	Batting Average	Frequency of hits per at-bat (H / AB)

Model Approach: Scout Score

Player	Team	AVG	RC	Scout_Score
Daniel Rogers	MAD	.472	21.08	.491
Blake Cavill	MOT	.301	28.71	.414
Ethan Surowiec	DUL	.387	73.70	.398
Dominic Kibler	KEN	.307	21.00	.390
Junior Barajas	WIR	.294	11.87	.384

Key Findings

Focus: Provides a conceptual framework for wood-bat evaluation rather than a final performance metric
Evaluation Scope: Prioritizes the qualitative mechanics of the wood-bat transition over quantitative hitter rankings
Contextual Tooling: Serves as a guide for scouting and player development - not a definitive grade of talent
Analytical Limits: Recognizes that while the framework tracks key variables, it is not a predictive measure of hitter success

Contact

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Sources

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